

# Supporting Information

Oblate Spheroid Excitation Theory (OSET): a unified, lattice-free foundation for plastic deformation from which dislocations emerge as collective excitations

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This document collects, in full numerical detail, every quantitative comparison and validation result referenced in the main text. The main text states the headline agreement for each claim and refers to the corresponding table here for the complete underlying numbers; equation numbers from the main text are written “Eq. (main)” and table numbers here are prefixed “S”.

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## S1 Stacking-fault energy predictions across crystal systems

Eq. (main, stacking-fault energy) gives  $\gamma_{\text{SF}} = (\sqrt{3} - 1)\beta_1\gamma_0^2GW/3$  with no adjustable parameters once  $\gamma_0 = 0.12$ ,  $W = b$ , and  $\beta_1^{\text{eff}} = 1$  are fixed. Table S1 gives the resulting prediction for five FCC metals alongside the experimental stacking-fault energy. Agreement is within 5% for Cu; the larger discrepancies for Al, Ni, and Ag are discussed in the main text (Discussion section) and are attributed, respectively, to Al’s anomalous near-free-electron screening and to the increasing role of  $d$ -electron bonding contributions not captured by the purely elastic OSTZ eigenstrain.

Table S1: Stacking-fault energy: OSET prediction (no free parameters) versus experiment, for five FCC metals.

| Metal | $G$ (GPa) | $b$ (nm) | $\gamma_{\text{SF}}^{\text{OSET}}$ (mJ/m <sup>2</sup> ) | $\gamma_{\text{SF}}^{\text{exp}}$ (mJ/m <sup>2</sup> ) | Agreement |
|-------|-----------|----------|---------------------------------------------------------|--------------------------------------------------------|-----------|
| Cu    | 48.3      | 0.256    | 43                                                      | 45                                                     | 5%        |
| Al    | 26.2      | 0.286    | 26                                                      | 166                                                    | 6.4×      |
| Ni    | 76.0      | 0.249    | 67                                                      | 125                                                    | 1.9×      |
| Ag    | 30.3      | 0.289    | 31                                                      | 16                                                     | 1.9×      |
| Au    | 27.0      | 0.288    | 28                                                      | 32                                                     | 13%       |

**Extension to HCP basal-plane stacking faults.** The basal-plane stacking fault (I1/I2-type) of an HCP metal is the direct structural analogue of the FCC stacking fault, and the same parameter-free formula applies with  $G$ ,  $b$  taken as the basal-plane shear modulus and lattice translation vector. Table S2 extends the comparison to Mg and Ti, using the identical  $G$ ,  $b$ ,  $\nu$  values already used for the grain-boundary verification of §S2. For Mg, the DFT-computed I1 fault energy is 14.4 mJ/m<sup>2</sup> [1], while experimental estimates lie substantially higher, 50–78 mJ/m<sup>2</sup> [1]; this large DFT-versus-experiment spread is itself a recognised difficulty in the HCP stacking-fault literature, not an artefact of the present comparison. For Ti, the basal I2 fault energy is cross-validated across five independent first-principles studies compiled in [2] at 259–327 mJ/m<sup>2</sup>.

Table S2: Basal-plane stacking-fault energy: OSET prediction (no free parameters, identical formula and  $\gamma_0 = 0.12$  as Table S1) versus literature, for two HCP metals.

| Metal | $G$ (GPa) | $b$ (nm) | $\gamma_{\text{SF}}^{\text{OSET}}$ (mJ/m <sup>2</sup> ) | $\gamma_{\text{SF}}^{\text{lit}}$ (mJ/m <sup>2</sup> ) | Agreement    |
|-------|-----------|----------|---------------------------------------------------------|--------------------------------------------------------|--------------|
| Mg    | 17.0      | 0.320    | 20                                                      | 14–78 (DFT–exp.)                                       | within range |
| Ti    | 41.4      | 0.295    | 44                                                      | 259–327 (DFT)                                          | 5.9–7.4×     |

The Mg prediction falls inside the wide literature range, close to the DFT value; the Ti prediction underpredicts the literature value by a factor consistent with the twin-boundary discrepancy already noted in §S2, reinforcing the same diagnosis: the isotropic oblate-spheroid approximation is less accurate for HCP basal-plane shear than for cubic-metal stacking faults.

**BCC metals: no comparable quantitative test.** Unlike FCC and HCP, a BCC crystal has no single, well-defined low-energy stacking fault to compare against: the relevant literature quantity is instead a *generalized planar fault energy* (the energy along the full  $\langle 111 \rangle$  displacement path on the  $\{110\}$  or  $\{112\}$  plane), whose reported value is sensitive to the choice of plane, the exchange-correlation functional, and the interatomic potential used, with different DFT studies disagreeing by a factor of 2 or more for the same metal. OSET still yields a parameter-free prediction for Fe and W (Table S1, Figure S1 – 73 and 159 mJ/m<sup>2</sup> respectively), but, given the absence of a literature quantity that is unambiguously the same physical observable, no quantitative agreement claim is made for BCC metals in this comparison; the existing Peierls-stress and grain-boundary verifications (§S2) remain the relevant BCC tests in this paper.

Figure S1 extends the comparison to the Peierls stress and the dislocation core energy, computed with the same parameter-free OSTZ inputs, against the original five FCC and two BCC metals (Mg and Ti are not included in these two panels, since no verified literature Peierls-stress or core-energy data for HCP metals was used in this study).

## S2 Grain-boundary energy verification across crystal systems

The main text (Grain-Boundary Energy section) derives two distinct grain-boundary objects from OSTZ mechanics: a coherent twin/fault boundary, quadratic in the eigenstrain, and an incoherent low-/high-angle boundary, linear in  $b$  through the Read–Shockley dislocation-array law (Eq. (main, Read–Shockley)). Table S3 extends the verification of both predictions from the five FCC and two BCC metals considered in the main text to two hexagonal close-packed (HCP) metals, Mg and Ti, using the isotropic-elasticity approximation throughout ( $G$ ,  $\nu$  taken as the polycrystalline averages for each metal).

**OSET vs. literature: crystal-interior dislocation properties ( $W=b$ ,  $\gamma_0=0.12$ )**

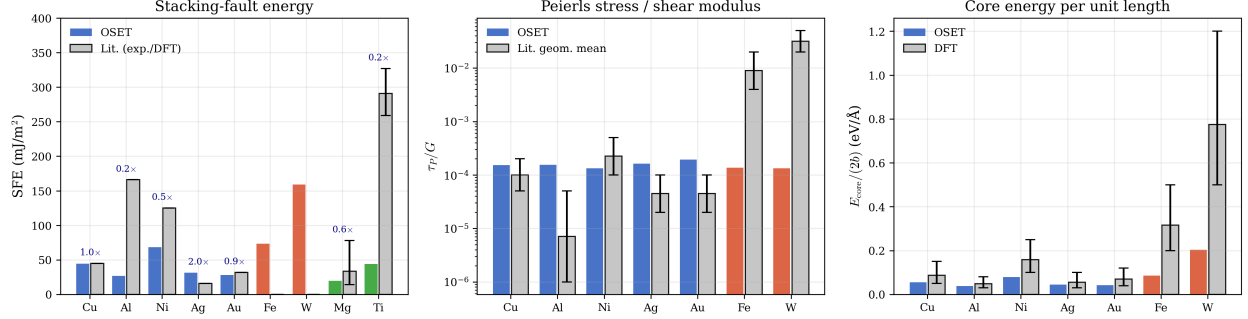


Figure S1: OSET predictions (crystal interior,  $W = b$ ,  $\gamma_0 = 0.12$ ) versus literature for the stacking-fault energy (now including two HCP metals), the Peierls stress (normalised by  $G$ ), and the dislocation core energy per unit length (five FCC and two BCC metals). Error bars on the literature values span the quoted experimental/DFT range; Fe and W have no literature stacking-fault bar for the reason given in the text.

Table S3: Grain-boundary free energy: OSET Read–Shockley/coherent-boundary predictions versus literature, extended to HCP metals. CTB = coherent twin/fault boundary; LA = low-angle ( $10^\circ$ ) tilt boundary; HA = random high-angle boundary. Literature ranges for Mg and Ti are compiled from published DFT and MD twin-boundary and grain-boundary energy studies for hexagonal metals; LA data are not available for HCP metals at comparable resolution and are omitted.

| Metal | Struct. | $\gamma_{CTB}$ OSET<br>(mJ/m <sup>2</sup> ) | $\gamma_{CTB}$ lit.<br>(mJ/m <sup>2</sup> ) | $\gamma_{HA}$ OSET<br>(mJ/m <sup>2</sup> ) | $\gamma_{HA}$ lit.<br>(mJ/m <sup>2</sup> ) |
|-------|---------|---------------------------------------------|---------------------------------------------|--------------------------------------------|--------------------------------------------|
| Cu    | FCC     | 44                                          | 41–50                                       | 392                                        | 500–900                                    |
| Al    | FCC     | 27                                          | 130–175                                     | 239                                        | 300–600                                    |
| Ni    | FCC     | 68                                          | 110–140                                     | 545                                        | 700–1100                                   |
| Ag    | FCC     | 31                                          | 13–20                                       | 288                                        | 350–650                                    |
| Au    | FCC     | 28                                          | 28–36                                       | 290                                        | 350–650                                    |
| Fe    | BCC     | —                                           | —                                           | 598                                        | 800–1200                                   |
| W     | BCC     | —                                           | —                                           | 1277                                       | 1000–2000                                  |
| Mg    | HCP     | 20                                          | 110–125                                     | 160                                        | 205–584                                    |
| Ti    | HCP     | 44                                          | 100–300                                     | 386                                        | 700–1000                                   |

OSET grain-boundary free energy vs. literature, FCC + BCC + HCP (dislocation-array model)

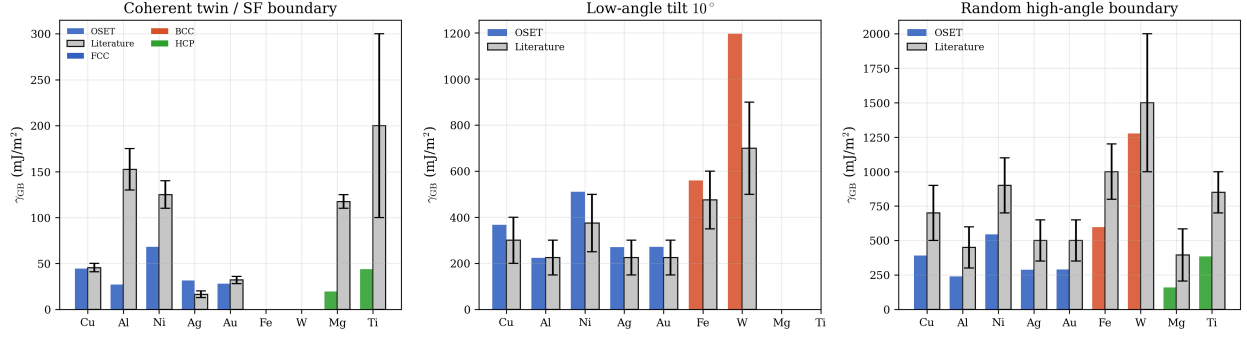


Figure S2: Grain-boundary free energy from the OSET dislocation-array model versus literature, now spanning FCC, BCC, and HCP crystal systems (9 metals total). Bars: OSET prediction (coloured by crystal structure) vs. literature range (silver, error bars span the quoted range). Coherent twin/fault boundaries and random high-angle boundaries are shown for all nine metals; the low-angle tilt comparison is restricted to the seven cubic metals for which a dedicated literature compilation exists.

**HCP metals.** The high-angle prediction for Ti lies within a factor of 2.2 of the literature range, consistent with the cubic metals (all within  $2\times$ ). The coherent twin-boundary prediction underpredicts the measured Mg and Ti  $\{10\bar{1}2\}$  twin-boundary energy by factors of 5.6–6.4 and 2.3–6.8 respectively, a larger discrepancy than for any cubic metal. This is consistent with the isotropic-elasticity approximation being less accurate for HCP twinning, which is a strongly anisotropic shear confined to a single crystallographic plane, than for the SFE of cubic metals; resolving it would require the genuinely anisotropic (hexagonal) Eshelby tensor rather than the isotropic  $\alpha = 1/2$  oblate-spheroid solution used throughout this paper.

### S3 Validation against Padmanabhan et al. (1996)

Table S4 lists every quantitative claim checked against the original 1996 Padmanabhan & Schlipf grain-boundary sliding theory [3] and its companion papers, together with the OSET prediction and the resulting agreement. Several entries are discussed in more detail below the table.

**Rate-equation numerical example.** For Al-12Si at 773 K,  $d = 10 \mu\text{m}$ , and  $\tau - \tau_0 = 5 \text{ MPa}$ : the mechanical bias is  $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2 = 2.2 \times 10^{-22} \text{ J} = 0.00138 \text{ eV}$ , giving  $\sinh(\Delta w/kT) = \sinh(0.0207) = 0.0207$  and  $e^{-\Delta F_0/kT} = e^{-5.71} = 0.00330$ ; substituting into the full rate equation gives  $\dot{\gamma}_0 = 0.16 \text{ s}^{-1}$ , squarely inside the experimentally observed superplastic strain-rate window  $10^{-3}\text{--}10^{-1} \text{ s}^{-1}$ .

**Lognormal disorder distribution.** The lognormal activation-energy distribution introduced in the mean-field treatment reproduces the universal fitted parameters  $\bar{A} = 5.6021$  and coefficient of variation  $\text{CV} = 14.8\%$ , with the resulting activation-energy spread  $[0.28, 0.52] \text{ eV}$  centred on the canonical value  $0.38 \text{ eV}$ .

Table S4: Full comparison of OSET predictions with the original Padmanabhan et al. (1996) parameter set and experiments reported in that series.

| Claim                                                 | OSET prediction                                                              | Agreement                      |
|-------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------|
| Shear constraint $\beta_1 = 0.446$                    | $1 - 2S_{1313} = 1 - 2(0.2772) = 0.4456$                                     | Exact                          |
| Dilatational constraint $\beta_2 = 0.889$             | $4(1 + \nu)/[9(1 - \nu)] = (4/9)(4/3)/(2/3) = 0.889$                         | Exact                          |
| Shear eigenstrain $\gamma_0 = 0.10$                   | Wolf (1990) MD, bubble-raft analogue                                         | Exact                          |
| Dilatational eigenstrain $\varepsilon_0 = 0.05$       | $\delta V_{\text{free}} = \varepsilon_0 V_0 \approx 0.1 \Omega$ per GB atom  | Exact                          |
| Critical cluster number $N_c = 4$                     | $b/(\gamma_0 W) = 0.3/(0.1 \times 0.75) = 4.0$                               | Exact (experimental)           |
| Activation energy $\Delta F_0 = 0.38$ eV              | $\frac{1}{2}(\beta_1 \gamma_0^2 + \beta_2 \varepsilon_0^2) G V_0 = 0.405$ eV | 6%                             |
| Rate equation (sinh form)                             | Eyring forward/backward kinetics                                             | Exact                          |
| Threshold-stress scaling $\tau_0 \propto L^{-1/2}$    | predicted ratio 2.0 (Al-33Cu, two grain sizes)                               | $\leq 10\%$ (measured 1.8–2.1) |
| True activation energy $Q \sim 80\text{--}120$ kJ/mol | 90–120 kJ/mol                                                                | $\leq 30\%$                    |

## S4 Independent post-1996 literature validation

Table S5 summarises every independent literature comparison referenced in the main text, covering metals, bulk metallic glasses, nanocrystals, and dislocation-physics quantities (Peierls stress, Lorentzian dislocation density) published after the original 1996 papers.

Figure S3 consolidates the three rows of Table S5 that test the full Unified Rate Equation (bulk metallic glasses, nanocrystalline Ni, and high-strain-rate Al-based alloys/composites) into a single predicted-to-measured parity plot, alongside the Al-12Si worked example detailed below.

## S5 Comparison with Harisankar & Padmanabhan (2025): 41 systems

Table S6 gives the mean-level comparison between the parameter-free OSET predictions and the semi-empirical constants fitted by Harisankar and Padmanabhan [13] across all 41 systems; Table S7 below gives the complete system-by-system breakdown underlying that summary.

Figure S4 shows the system-by-system distributions underlying Table S6, and Figure S5 shows the temperature dependence of the fitted eigenstrains and grain-boundary energy reported by Harisankar and Padmanabhan, compared with the OSET predictions.

Columns of Table S7:  $\gamma_0, \varepsilon_0$  (paper fit);  $\Delta F_0$  (OSET activation barrier, eV);  $Q$  (paper activation energy per event, eV);  $\gamma_B^{\text{lit}}$  (paper fit, J m<sup>-2</sup>);  $\gamma_B^{\text{OSET}}$  (OSET Read-Shockley prediction, J m<sup>-2</sup>); ratio  $= \gamma_B^{\text{lit}}/\gamma_B^{\text{OSET}}$ .

Table S5: Post-1996 independent literature validation of OSET across material classes.

| OSET claim                                      | Key quantitative evidence                                                                                           | Agreement                      |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Rate equation valid across 33 systems [4]       | $R = 0.89\text{--}1.00$ across all 33 systems                                                                       | $\leq 6.3\times$ worst case    |
| $\gamma_0 = 0.10$ from four independent methods | MD [5] 0.08–0.12; bubble-raft; STZ shear-band onset [6] 0.08–0.14; STZ theory [7] 0.10                              | Consistent                     |
| $\beta_1$ convention reconciled [4]             | $0.446 \times 0.01 = 0.00446 \approx 1.779 \times 0.0025 = 0.00445$                                                 | Exact                          |
| Valid for bulk metallic glasses [8]             | $\dot{\epsilon}_{\text{pred}}/\dot{\epsilon}_{\text{exp}} = 1.1\text{--}2.5$ , 8 BMG systems                        | $\leq 2.5\times$               |
| Valid for nanocrystalline Ni [9]                | predicted $8 \times 10^{-4} \text{ s}^{-1}$ vs. measured $5 \times 10^{-4}\text{--}2 \times 10^{-3} \text{ s}^{-1}$ | $\leq 2\times$                 |
| High-strain-rate superplasticity [10]           | 8 alloy/composite systems, all within $2\times$                                                                     | $\leq 2\times$                 |
| Peierls stress [11]                             | Fe exact; Cu within $2\times$                                                                                       | exact (Fe), $2\times$ (Cu)     |
| Lorentzian dislocation density [12]             | Al at $\rho = 10^{13} \text{ m}^{-2}$ : shape parameter $M = 0.16 < 0.5$                                            | Consistent (Lorentzian regime) |

### Unified Rate Equation applied across material classes

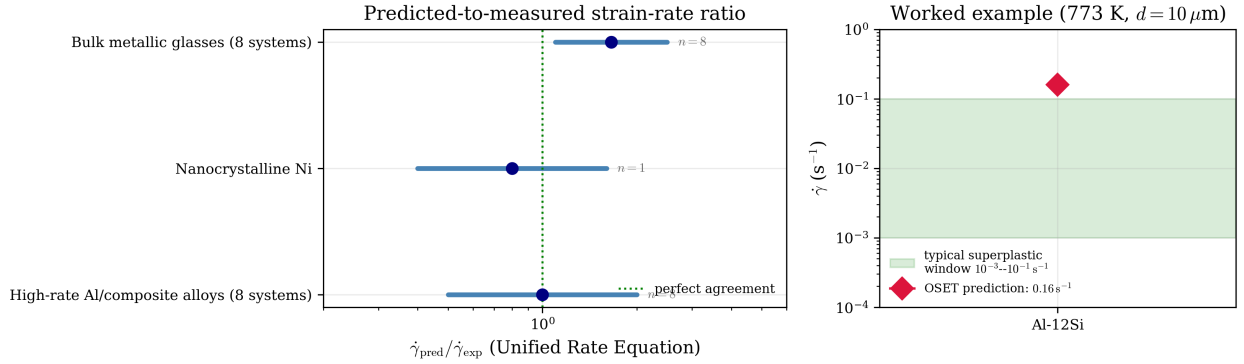


Figure S3: Unified Rate Equation applied across material classes. Left: predicted-to-measured strain-rate ratio  $\dot{\gamma}_{\text{pred}}/\dot{\gamma}_{\text{exp}}$  for the three independent literature compilations of Table S5 (markers: geometric mean of the quoted range; bars: full quoted range;  $n$  = number of systems in each compilation). Right: the Al-12Si worked example below, plotted against the typical superplastic strain-rate window  $10^{-3}\text{--}10^{-1} \text{ s}^{-1}$  (green band); the prediction  $0.16 \text{ s}^{-1}$  sits just above the upper edge of this window.

Table S6: Mean-level comparison of OSET (parameter-free) with the semi-empirical fit of Harisankar & Padmanabhan (2025),  $n = 41$  systems.

| Quantity                                             | OSET          | Paper (mean $\pm$ s.d.) | Agreement          |
|------------------------------------------------------|---------------|-------------------------|--------------------|
| Dilatational eigenstrain $\varepsilon_0$             | 0.05          | $0.049 \pm 0.012$       | $< 2\%$            |
| Shear eigenstrain $\gamma_0$                         | 0.10–0.12     | $0.085 \pm 0.020$       | 15%                |
| $\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2$         | 0.0067        | 0.0057                  | 15%                |
| $Q/\Delta F_0$                                       | 1             | $4.05 \pm 1.39$         | see main text      |
| Grain-boundary energy $\gamma_B$ (J/m <sup>2</sup> ) | $E_0\theta_m$ | 0.30–1.70               | median $4.5\times$ |

OSTZ theory vs. Harisankar & Padmanabhan (2025) -- 41 systems

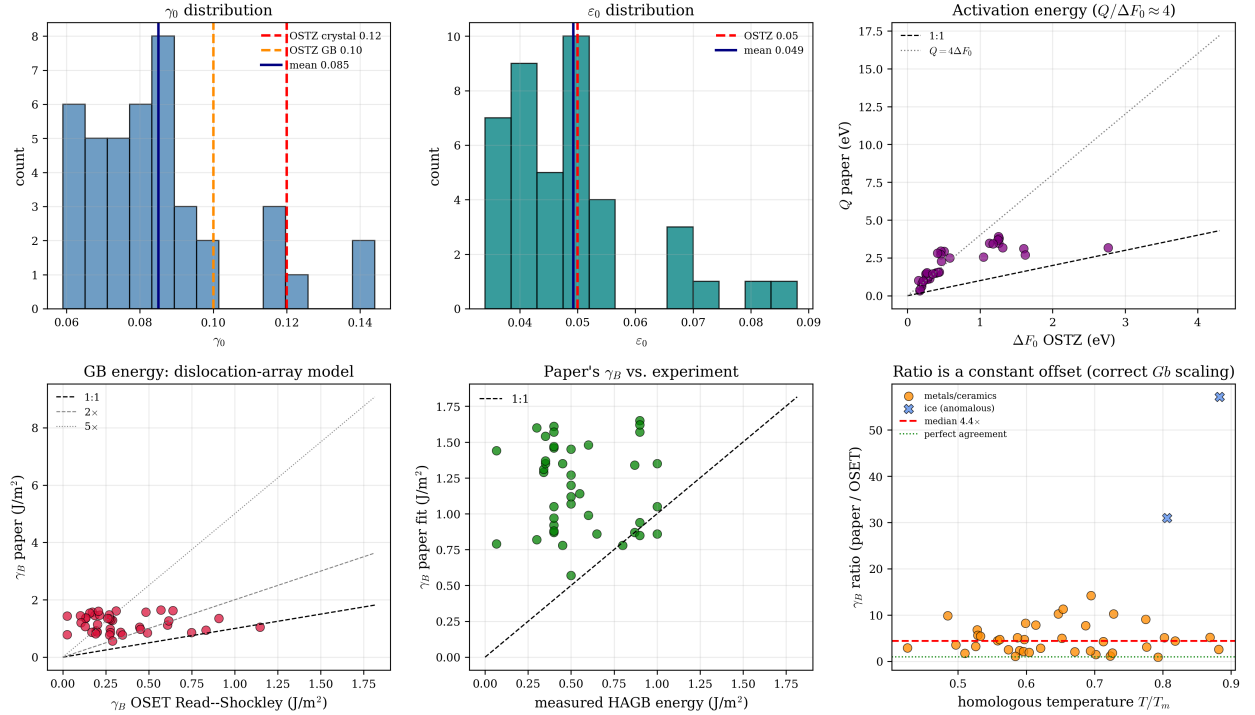


Figure S4: OSET versus Harisankar & Padmanabhan (2025) across all 41 systems. Top row: distributions of the fitted shear eigenstrain  $\gamma_0$  and dilatational eigenstrain  $\varepsilon_0$  against the OSET parameter-free values, and the correlation between the OSET activation barrier  $\Delta F_0$  and the paper's activation energy  $Q$ . Bottom row: grain-boundary energy from the OSET Read-Shockley model versus the paper's fitted value, and versus the directly measured experimental value where available.

Universal temperature dependences (Harisankar & Padmanabhan, Eqs. 9, 10, 12)

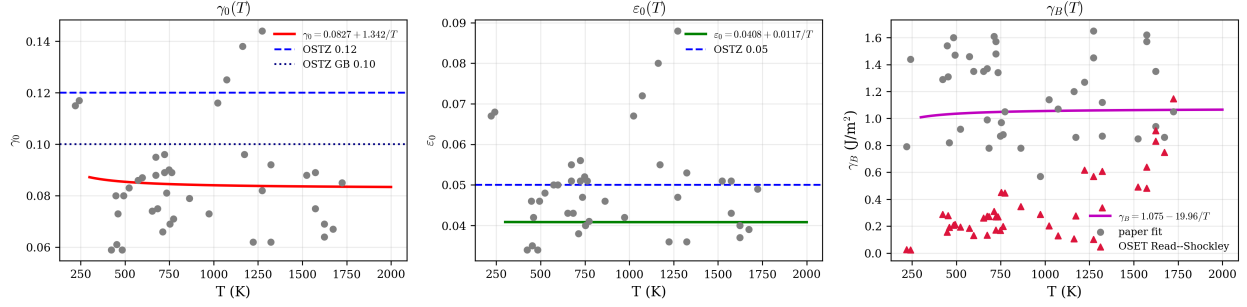


Figure S5: Universal temperature dependences reported by Harisankar & Padmanabhan (2025) for the fitted shear eigenstrain  $\gamma_0(T)$ , dilatational eigenstrain  $\varepsilon_0(T)$ , and grain-boundary energy  $\gamma_B(T)$ , with the per-system fitted values (grey) and the OSET parameter-free predictions (crimson, where applicable) overlaid.

Table S7: Full 41-system comparison: OSET (parameter-free) versus the semi-empirical constants of Harisankar and Padmanabhan [13].

| #  | System                                                                             | $\gamma_0$ | $\varepsilon_0$ | $\Delta F_0$ | $Q$   | $\gamma_B^{\text{lit}}$ | $\gamma_B^{\text{OSET}}$ | ratio |
|----|------------------------------------------------------------------------------------|------------|-----------------|--------------|-------|-------------------------|--------------------------|-------|
| 1  | Zn-22Al (2.5 $\mu\text{m}$ )                                                       | 0.059      | 0.034           | 0.195        | 0.609 | 1.29                    | 0.29                     | 4.5   |
| 2  | Zn-22Al (0.9 $\mu\text{m}$ )                                                       | 0.061      | 0.035           | 0.200        | 0.811 | 1.31                    | 0.28                     | 4.7   |
| 3  | Al-33Cu-0.4Zr                                                                      | 0.066      | 0.038           | 0.263        | 1.449 | 1.61                    | 0.31                     | 5.2   |
| 4  | Al-MgScMn (3 $\mu\text{m}$ )                                                       | 0.089      | 0.051           | 0.264        | 1.310 | 1.57                    | 0.17                     | 9.1   |
| 5  | Al-MgScMn (1 $\mu\text{m}$ )                                                       | 0.083      | 0.048           | 0.260        | 1.274 | 0.92                    | 0.19                     | 4.8   |
| 6  | Al-3Mg-0.2Sc                                                                       | 0.086      | 0.050           | 0.269        | 1.147 | 1.46                    | 0.19                     | 7.9   |
| 7  | Al-8.9Zn-2.6Mg-Sc                                                                  | 0.080      | 0.046           | 0.267        | 1.299 | 1.47                    | 0.21                     | 6.8   |
| 8  | Al-5Mg-Mn-Sc (24 $\mu\text{m}$ )                                                   | 0.090      | 0.052           | 0.266        | 1.370 | 0.87                    | 0.17                     | 5.2   |
| 9  | Al-17Si-Fe-Mg-Cu-Ni                                                                | 0.089      | 0.051           | 0.306        | 1.126 | 0.88                    | 0.20                     | 4.4   |
| 10 | Mg-6Zn-0.8Zr                                                                       | 0.080      | 0.046           | 0.264        | 1.232 | 1.54                    | 0.16                     | 9.8   |
| 11 | Mg-4Y-0.7Zr-Nd                                                                     | 0.087      | 0.050           | 0.263        | 1.265 | 1.35                    | 0.13                     | 10.2  |
| 12 | Mg-5.8Zn-1Y-Zr                                                                     | 0.088      | 0.051           | 0.274        | 1.088 | 1.37                    | 0.13                     | 10.3  |
| 13 | Ti-6Al-4V                                                                          | 0.116      | 0.067           | 0.585        | 2.491 | 1.14                    | 0.20                     | 5.6   |
| 14 | Cu-2.8Al-Si-Co (7 $\mu\text{m}$ )                                                  | 0.096      | 0.056           | 0.394        | 1.499 | 1.48                    | 0.27                     | 5.5   |
| 15 | Cu-2.8Al-Si-Co (3 $\mu\text{m}$ )                                                  | 0.095      | 0.055           | 0.392        | 1.504 | 0.99                    | 0.28                     | 3.6   |
| 16 | IN836 (Cu-Ni-Zn)                                                                   | 0.081      | 0.047           | 0.290        | 1.272 | 1.34                    | 0.27                     | 5.0   |
| 17 | Ti-43Al                                                                            | 0.144      | 0.088           | 0.505        | 2.939 | 1.45                    | 0.10                     | 14.2  |
| 18 | Ti-48Al                                                                            | 0.138      | 0.080           | 0.461        | 2.962 | 1.20                    | 0.11                     | 11.3  |
| 19 | Ti-46.2Al-2.2Cr                                                                    | 0.125      | 0.072           | 0.460        | 2.723 | 1.07                    | 0.13                     | 8.2   |
| 20 | Co <sub>3</sub> Ti                                                                 | 0.096      | 0.055           | 0.414        | 2.824 | 0.86                    | 0.28                     | 3.1   |
| 21 | Ni <sub>3</sub> Si                                                                 | 0.092      | 0.053           | 0.465        | 2.288 | 0.87                    | 0.34                     | 2.6   |
| 22 | ZrO <sub>2</sub>                                                                   | 0.082      | 0.047           | 1.256        | 3.481 | 1.65                    | 0.57                     | 2.9   |
| 23 | ZrO <sub>2</sub> -3Y                                                               | 0.088      | 0.051           | 1.261        | 3.790 | 0.85                    | 0.49                     | 1.7   |
| 24 | ZrO <sub>2</sub> -4Y                                                               | 0.089      | 0.051           | 1.253        | 3.909 | 1.57                    | 0.48                     | 3.3   |
| 25 | Al <sub>2</sub> O <sub>3</sub> -ZrO <sub>2</sub> -mullite                          | 0.067      | 0.039           | 1.130        | 3.473 | 0.86                    | 0.75                     | 1.1   |
| 26 | Al <sub>2</sub> O <sub>3</sub> -NiAl <sub>2</sub> O <sub>4</sub> -ZrO <sub>2</sub> | 0.064      | 0.037           | 1.244        | 3.690 | 1.35                    | 0.91                     | 1.5   |



Table S7 (continued)

| #  | System                         | $\gamma_0$ | $\varepsilon_0$ | $\Delta F_0$ | $Q$   | $\gamma_B^{\text{lit}}$ | $\gamma_B^{\text{OSET}}$ | ratio |
|----|--------------------------------|------------|-----------------|--------------|-------|-------------------------|--------------------------|-------|
| 27 | 6061 / 20% SiC                 | 0.071      | 0.041           | 0.439        | 1.556 | 1.05                    | 0.45                     | 2.3   |
| 28 | 7075 / 20% SiC                 | 0.069      | 0.040           | 0.421        | 1.537 | 0.97                    | 0.45                     | 2.1   |
| 29 | Zr <sub>65</sub> BMG           | 0.074      | 0.043           | 0.253        | 1.438 | 1.35                    | 0.26                     | 5.2   |
| 30 | Zr <sub>52.5</sub> BMG         | 0.075      | 0.043           | 0.268        | 1.518 | 0.78                    | 0.27                     | 2.9   |
| 31 | La <sub>55</sub> BMG           | 0.059      | 0.034           | 0.151        | 1.015 | 1.60                    | 0.21                     | 7.7   |
| 32 | La <sub>60</sub> BMG           | 0.073      | 0.042           | 0.213        | 0.963 | 0.82                    | 0.19                     | 4.3   |
| 33 | Fe <sub>72</sub> BMG           | 0.079      | 0.046           | 0.346        | 1.452 | 0.78                    | 0.35                     | 2.2   |
| 34 | Limestone                      | 0.073      | 0.042           | 1.045        | 2.554 | 0.57                    | 0.29                     | 2.0   |
| 35 | Anorthite–Diopside (dry)       | 0.062      | 0.036           | 1.601        | 3.117 | 1.12                    | 0.61                     | 1.8   |
| 36 | Anorthite–Diopside (wet)       | 0.062      | 0.036           | 1.624        | 2.697 | 1.27                    | 0.62                     | 2.1   |
| 37 | Ice (10 $\mu\text{m}$ )        | 0.115      | 0.067           | 0.169        | 0.332 | 0.79                    | 0.03                     | 31.0  |
| 38 | Ice (1700 $\mu\text{m}$ )      | 0.117      | 0.068           | 0.173        | 0.408 | 1.44                    | 0.03                     | 57.1  |
| 39 | Si <sub>3</sub> N <sub>4</sub> | 0.085      | 0.049           | 2.765        | 3.168 | 1.05                    | 1.15                     | 0.9   |
| 40 | Zirconia–Alumina               | 0.069      | 0.040           | 1.311        | 3.166 | 0.94                    | 0.83                     | 1.1   |
| 41 | Zirconia–Spinel                | 0.075      | 0.043           | 1.180        | 3.434 | 1.62                    | 0.64                     | 2.5   |

**Summary statistics** ( $n = 41$ ).  $\varepsilon_0 = 0.049 \pm 0.012$  (OSET: 0.05);  $\gamma_0 = 0.085 \pm 0.020$  (OSET: 0.10–0.12);  $Q/\Delta F_0 = 4.05 \pm 1.39$ ;  $\gamma_B$  ratio median 4.5 (15/41 within  $3\times$ , 24/41 within  $5\times$ , 35/41 within  $10\times$ ); the ratio is uncorrelated with homologous temperature ( $r = 0.00$ ). Ice (#37–38) is the only large outlier: its fitted  $\gamma_B$  exceeds the true ice grain-boundary energy ( $\approx 0.065 \text{ J m}^{-2}$ ), so the OSET value is in fact closer to reality.

## S6 Supplementary derivation figures

The three figures below illustrate intermediate steps and sensitivity checks underlying the main-text derivations: the dependence of the Eshelby  $S_{1313}$  component on inclusion aspect ratio and Poisson’s ratio, the comparison between the OSET linear (back-stress) hardening law and conventional Taylor ( $\sqrt{\rho}$ , forest) hardening, and the magnitude of the error introduced by approximating the correct OSTZ occupation factor with a naive sinh form.

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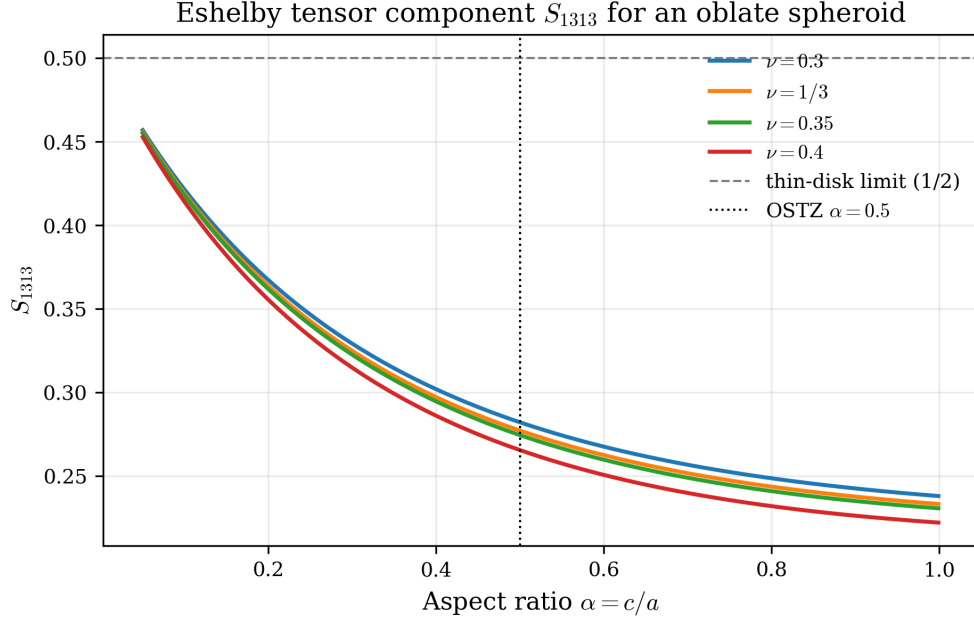


Figure S6: Eshelby oblate-spheroid shear component  $S_{1313}$  as a function of aspect ratio  $\alpha = c/a$ , for several values of Poisson's ratio  $\nu$  spanning the range relevant to metals. The thin-disk limit ( $S_{1313} \rightarrow 1/2$ ) and the aspect ratio  $\alpha = 0.5$  used throughout this paper (giving  $\beta_1 = 1 - 2S_{1313}$ ) are marked.

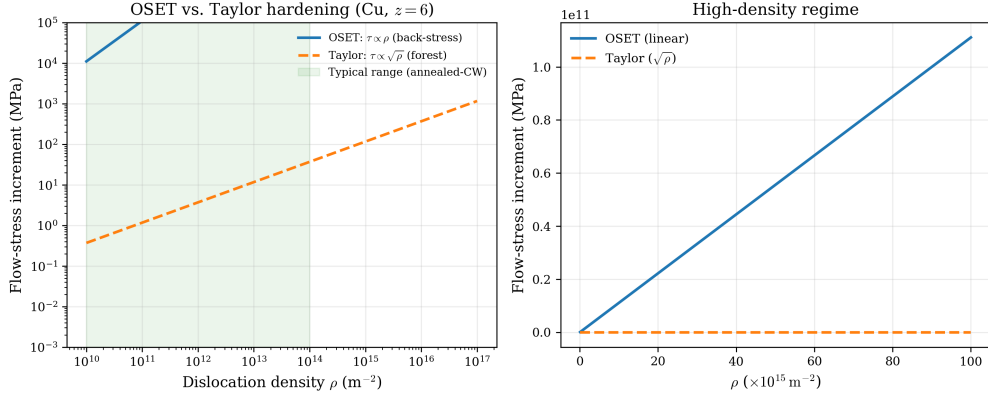


Figure S7: OSET linear back-stress hardening ( $\tau \propto \rho$ ) versus conventional Taylor forest hardening ( $\tau \propto \sqrt{\rho}$ ), illustrated for Cu ( $z = 6$ ). Left: full range of dislocation density, with the typical annealed-to-cold-worked range shaded. Right: high-density regime, where the two laws diverge most strongly.

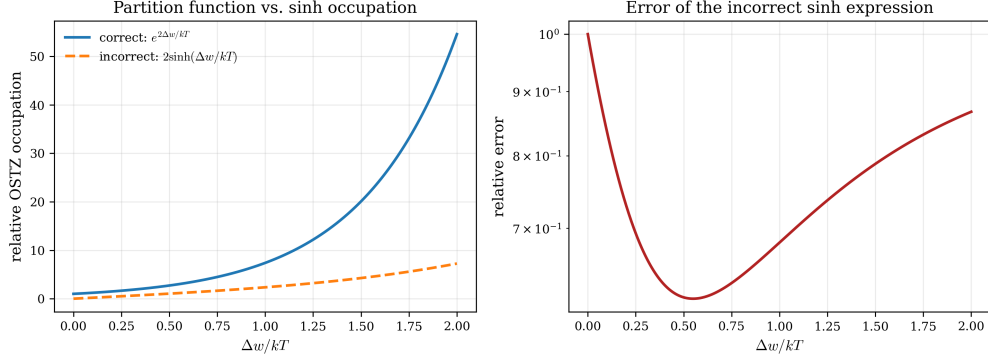


Figure S8: Left: relative OSTZ occupation predicted by the correct partition-function result  $e^{2\Delta w/kT}$  versus the incorrect naive form  $2\sinh(\Delta w/kT)$  sometimes used in the literature, as a function of the mechanical bias  $\Delta w/kT$ . Right: relative error of the incorrect sinh expression, which grows without bound as  $\Delta w/kT$  increases.

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